3.2 AUDIO SYSTEM

Some radio systems available on the 2006 TJ communicate on the PCI Bus. They use the bus for three reasons. The first is to communicate trouble codes, second is to receive dimming information, and the third is to receive cabin equalization information. The audio system is available in a 4 speaker base system and a 7 speaker system with an external subwoofer.

When troubleshooting output shorts or "output" error messages, the following applies:

On radios without an external amplifier, the term output refers to the path between the radio and the speaker. This type of circuit can be monitored all the way through the speaker connections by the radio assembly. When the radio displays a shorted output DTC with this type of system, the speaker, radio, or wiring could be at fault.

On radios with an external amplifier, the term "output" refers to the circuit between the radio connector and the amplifier. The radio is capable of monitoring only this portion and can tell nothing about the circuit between the amplifier and the speakers. Consequently, a shorted output DTC on this type of system would only refer to this circuit. A faulty speaker could not cause this DTC.

3.2.1 SATELLITE RADIO (SDARS)

To use the satellite radio, the radio must be in satellite mode. The satellite antenna must have an unobstructed view of the satellite. The vehicle must be outside to receive proper reception. Poor reception may be experienced in large down town areas where tall buildings are present. The satellite network is still under development, and this could be the cause of poor reception in certain areas. Use a known good system to verify network performance in these areas.

Once in satellite mode, the radio head is used to change stations. The Radio must be in Satellite mode to communicate to the Satellite Receiver Module with the DRBIII®. The Radio head communicates to the Satellite Receiver Module via the PCI bus and the private Audio bus. The Audio bus functions just like the PCI bus, but its use is restricted to the Radio and the Satellite Receiver Module.

The Satellite Receiver Module processes signals it receives from the satellite antenna. That information is then sent to the Radio where it is processed and sent out to the speakers. The Satellite Receiver Module has the ability to detect and store fault code information, which may be retrieved with the DRBIII®. Refer to the service manual for additional information on the SDAR system and its related components.

3.3 ELECTROCHROMIC COMPASS/TEMPERATURE MIRROR WITH LIGHTS

DESCRIPTION

The optional self-dimming Electrochromic Compass/Temperature Mirror has a vacuum fluorescent (VF) display that is integrated into the rear view mirror. The Compass/Temp Mirror includes the compass/temperature display and two map/reading lamps. This display provides the outside temperature and one of eight compass headings to indicate the direction the vehicle is facing. The Compass/Temp Mirror displays the compass heading and the outside temperature at the same time. The Ambient Temperature Sensor monitors the outside temperature and is hardwired to the Compass/Temp Mirror.

BUTTON OPERATION

The Compass/Temp Mirror incorporates 3 (mode/lamp) buttons to access and control various functions.

Left Button

- Press and Release
  - > Toggles Left Map Light On/Off
- Press w/ Right Switch for more than 5 seconds and Release
  - > Toggles Electrochromic Status On/Off -automatic On with each ignition cycle
  - > On mode is indicated by green status LED next to right button

Center Button

- Press for more than 3 seconds and Release
  - > Cycle through display status
    - – Compass / Temperature (Fahrenheit) - °F mode stored to memory
    - – Compass / Temperature (Celsius) - °C mode stored to memory
    - – Display Off - off mode stored to memory
- Press and Hold for 3-6 seconds
  - > Activate zone variance mode (adjust compass for true north)
    - – ″Z″ and the currently programmed zone (1-15) will flash in display window
    - – Each press of the center button will increment to the next zone
    - – When proper zone is selected, wait 5 seconds and mirror display returns to comp/temp (see variance zone map for proper zone number setting)
- Press and Hold for more than 6 seconds
  - > Activate compass calibration
    - – ″CAL″ shows in display window until compass is calibrated (verify correct variance zone